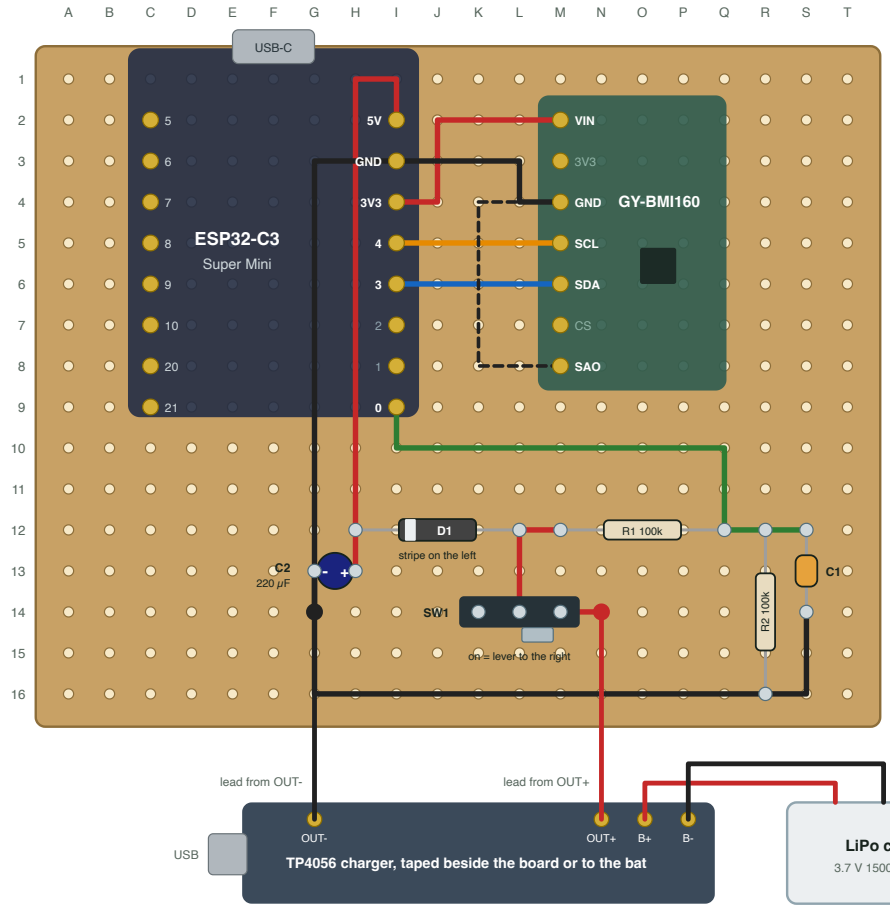
Build it on the zero PCB

Zero PCB layout, seen from the component side

One arrangement that works. Each dot is a hole, 2.54 mm from the next. The wires run on the solder side and must be insulated.



Wires, pad to pad

I4 to M2	3V3 to VIN
I3 to M4	GND to GND
I5 to M5	GPIO4 to SCL
I6 to M6	GPIO3 to SDA
M8 to M4	SAO to GND
N14 to M14	charger OUT+ to the switch
L14 to L12	switch to the diode
H12 to I2	diode to 5V
G14 to I3	charger OUT- to GND
H13 to H12	C2 plus to the 5V line
G13 to G14	C2 minus to GND

Only for stage 7

L12 to M12	switch output to R1
Q12 to R12	R1 to R2
R12 to S12	R2 to C1
Q12 to I9	R1 and R2 to GPIO0
S14 to R16	C1 to the ground end of R2
R16 to G14	that ground end to OUT-

Parts

```
D1 H12 (stripe) and L12
SW1 K14, L14, M14
C2 H13 (+) and G13 (−)
R1 M12 and Q12
R2 R12 and R16
C1 S12 and S14
OUT+ lead N14
OUT− lead G14
```

The hole positions assume this pin order: 5V, GND, 3V3, 4, 3, 2, 1, 0 down the right side of the ESP32, and VIN, 3V3, GND, SCL, SDA, CS, SAO on the sensor.
If your boards are printed in another order, keep the label-to-label connections of the table in the README and move the wires to suit.
The script that drew this checked it: the layout makes all 8 on-board connections of the circuit, joins nothing else, and no wire runs through another connection's pad.
Some wires cross each other, which is why they must be insulated. The two leads from the charger are soldered at N14 and G14 and leave the board there.

On a zero PCB every hole is a separate pad. Nothing is joined until you join it with solder or wire. The drawing shows one arrangement that works, seen from the side the parts sit on. The wires run on the other side, pad to pad.

The script that drew the layout also checked it: it makes every connection of the circuit, joins nothing else, and no wire runs through the pad of another connection. That does not make your soldering right. The multimeter checks below do.

You do not have to solder everything at once. Build in three sessions, and test after each.

Session 1: before stage 2

The two boards and the five sensor wires.

1. **Solder the pin strips to the two boards** if they came loose, pins pointing down. Push the strips into a breadboard or the female headers first, lay the board on top, and solder. That keeps the pins straight.

2. **Place the female headers.** For the ESP32: two 8-pin strips in columns C and I, rows 2 to 9. For the sensor: one 7-pin strip in column M, rows 2 to 8. The USB-C end of the ESP32 points to the edge of the board, so the cable can reach it.
3. **Plug both boards in before soldering anything**, to check the fit.
4. **Solder every header pin** on the underside.
5. **Add the five wires** on the underside:

From	To	Colour in the drawing	Connects
I4	M2	red	`3V3` to `VIN`
I3	M4	black	`GND` to `GND`
I5	M5	orange	`GPIO4` to `SCL`
I6	M6	blue	`GPIO3` to `SDA`
M8	M4	black, dashed	`SAO` to `GND`

6. **Check with the multimeter before plugging in USB.** Set it to continuity, the setting that beeps.
   - It must beep between the two ends of each wire in the table.
   - It must not beep between `3V3` and `GND`. If it does, there is a short. Find it before any power is connected.
   - It must not beep between `SCL` and `SDA`, or between any pad and its neighbour.

Now do stages 2 to 5.

Session 2: before stage 6

The switch, the diode, C2 and the two leads from the charger. Read the README of stage 6 first: it has the checks that keep the battery safe.

Part	Holes	Notes
D1, the diode	H12 and L12	**The striped end goes in H12.** That end leads to the `5V` pin.
SW1, the switch	K14, L14, M14	The middle pin is L14.
C2, 220 µF electrolytic	H13 and G13	**The long leg (+) goes in H13.** The short leg, on the side with the stripe, goes in G13. Bend it over to lie flat if it stands too tall.
Lead from the charger's `OUT+`	N14	Red wire.
Lead from the charger's `OUT−`	G14	Black wire.

From	To	Colour	Connects
N14	M14	red	`OUT+` to the switch
L14	L12	red	switch to the diode

From	To	Colour	Connects
H12	I2	red	diode to `5V`. The wire runs up under the ESP32 and round the top of its header.
G14	I3	black	`OUT−` to `GND`. This one also runs under the ESP32.
H13	H12	red	C2's plus lead to the `5V` line. A blob of solder across the two holes is enough.
G13	G14	black	C2's minus lead to `GND`. A blob of solder is enough here too.

No diode yet? Use the layout without the diode instead. Leave D1 out and solder a wire from L12 to H12 in its place:

From	To	Colour	Connects
L12	H12	red	a wire where the diode was

Then **switch the battery off before plugging a USB cable into the ESP32**, every time. Without the diode, the USB cable would push 5 volts into the cell. make-changes.pdf has the details.

The charger module does not fit the holes of the zero PCB. Fix it beside the board, or to the bat, with foam tape, and keep its USB socket reachable.

Session 3, optional: before stage 7

Part	Holes
R1, 100 kilohm	M12 and Q12
R2, 100 kilohm	R12 and R16
C1, 100 nanofarad	S12 and S14

From	To	Colour	Connects
L12	M12	red	switch output to R1
Q12	R12	green	R1 to R2
R12	S12	green	R2 to C1
Q12	I9	green	the middle point to `GPIO0`
S14	R16	black	C1 to the ground end of R2
R16	G14	black	that ground end to `OUT−`

If your boards are printed differently

The hole positions assume this pin order: `5V, GND, 3V3, 4, 3, 2, 1, 0` down the right side of the ESP32 with its USB socket up, and `VIN, 3V3, GND, SCL, SDA, CS, SAO` on the sensor. Boards from different makers differ. If yours do, keep the label-to-label connections of the table in the README and move the wires to suit.

Good habits

- **Insulated wire only.** Several wires cross on the underside.
- **Short wires for power and ground.** Long thin wires to the battery are the usual cause of restarts during a swing.
- **Clean joints.** A good joint is shiny and shaped like a small cone. A dull ball sitting on the pad is not connected.
- **Look for bridges.** Hold the board against a light. Solder that joins two neighbouring pads shows up at once.
- **Cover the underside** with tape or a piece of foam when everything works, so nothing metal can touch it on the bat.